

1.

What is the worst-case time complexity for searching in a hash table with collisions?

- A. $O(1)$**
- B. $O(\log n)$**
- C. $O(n^2)$**
- D. $O(n)$**

Answer : D

2.

Balance Factor of a node is

- A. Height of Right SubTree - Height of Left SubTree**
- B. Height of Left SubTree - Height of Right SubTree**
- C. $\text{Max}(\text{height of left, height of right tree}) + 1$**
- D. -1**

Answer : B

3.

What is a cycle in a graph?

- A. A sequence of edges with no repetition**
- B. A path where the starting and ending nodes are the same**
- C. A disconnected component**
- D. A loop on a node**

Answer : B

4.

What is a complete graph ?

- A. Every vertex has no edges**
- B. Every vertex is connected to every other vertex**
- C. No cycles**
- D. Vertices are divided into 2 disjoint sets.**

Answer : D

5.

What is the load factor in a hash table?

- A. Number of collisions**
- B. Size of the hash table**
- C. Ratio of the number of elements to the size of the table**
- D. Number of keys**

Answer : C

6.

Which of the following is true about a Binary Search Tree ?

- I. A Tree where all the nodes are arranged in ascending Order**
 - II. A tree where each node has a degree 2**
 - III. A Tree where each node can have a degree 0,1 or 2**
 - IV. A tree where the left child is less than the root and the right child is greater or equal to the root.**
- A. Only II**
 - B. Only III**
 - C. Both II and IV**
 - D. Both III and IV**

Answer : D